

Robin Lange

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Robotics and Autonomous Systems | Perception, Mission Logic and Physical Testing

Education

Georgia Southern University

Aug. 2023–May 2027

B.S. Mechanical Engineering, Vehicle Design & Testing concentration

GPA: 4.00/4.00

President's List in every completed graded term; Honors College Scholar; Tau Beta Pi (elected as junior); Allen E. Paulson Scholar.

Research Experience

Institute for Human & Machine Cognition (IHMC)

May–Aug. 2026

National Center for Collaborative Autonomy (NCCA)

Collaborative Robotics Autonomy Intern

Pensacola, FL

- Built a Rust/Python autonomy codebase from scratch for two Boston Dynamics Spot robots, implementing pickup/drop-off sequencing, JAG decision logic, coordinate transforms and keyed communications.
- Used inverse kinematics and local perception for Spot pickups; integrated Zenoh, Motive tracking and MCAP/Foxglove. Added markers to a manually operated Husky AGV.

Georgia Southern University

Oct. 2023–present

Undergraduate Researcher, Intelligent Vehicles Navigation Research Laboratory

Statesboro, GA

- Integrated visual SLAM, mapping and behavior-tree navigation on Jetson/RealSense UAV hardware; evaluated localization in a 110-second cart trial (14.5 cm RMSE on valid reference segments).
- Developed a LiDAR hand-motion interface; coauthored two SAE papers. Building a MATLAB/AVL baseline for an in-progress adaptive-cruise-control thesis.

CARISSMA Institute of Safety in Future Mobility

July 1–Aug. 15, 2025

Technische Hochschule Ingolstadt

Vehicle Testing Research Intern

Ingolstadt, Germany

- Designed and built covering, quick-assembly and fold-away mechanisms for a soft truck-rear crash target, developing an earlier CAD/foam concept into a physical prototype.
- Tested coverings and corner reflectors with LiDAR and 4D radar; analyzed qualitative returns using team scripts.

Fraunhofer Institute for Transportation and Infrastructure Systems IVI

June 1–Aug. 11, 2024

Advanced Air Mobility research

Autonomous Systems and AI Intern

Ingolstadt, Germany

- Developed Mission AI decision and execution logic in Python/ROS 2 and py_trees, comparing state-machine and behavior-tree approaches.
- Verified gated flight transitions and emergency interruption in simulation.

Technical Skills

Python, Rust, MATLAB, C++; ROS 2/DDS, Zenoh, Protocol Buffers; inverse kinematics, coordinate transforms, behavior trees, visual SLAM, motion capture, LiDAR, SolidWorks, Git/Linux.

Selected Publications and Awards

SAE technical papers (coauthor): Vehicle-simulation calibration (2026); rule-based LiDAR lane keeping (2025).

2026: Third Place Undergraduate, Gulfstream Innovation, Engineering and Flight Excellence Awards; NCUR

Engineering Division Video Contest Awardee. **2025:** Second Place Undergraduate, Georgia Power Innovation Awards.